Performance Testing

Date	04 July 2026
Team ID	**AI Powered Debt Relief & Financial Recovery Platform**
Project Name	SWTID-2026-2662
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Apache JMeter
Type of Testing	Load Testing, Stress Testing
Target Module / API	User Login, AI Debt Analysis, Budget Recommendation, Financial Report Generation
Test Environment	Local System (Windows 11, Python Streamlit, Google Gemini API)
Test Date	04 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	User Login Performance	50	60	Users should log in successfully without delay
2	AI Debt Analysis Request	100	120	AI should generate debt analysis within 2 seconds
3	Budget Recommendation Generation	150	180	Personalized budget recommendations generated successfully

4	Financial Report Download	75	90	PDF report downloads without errors

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.42 sec	Pass	Fast response
2	Response Time (Max)	< 5 seconds	3.68 sec	Pass	Acceptable under heavy load
3	Throughput (Req/sec)	> 40 Req/sec	55 Req/sec	Pass	Good request handling
4	Error Rate	< 1%	0.30%	Pass	Very few errors
5	CPU Utilization	< 80%	65%	Pass	Stable CPU usage
6	Memory Utilization	< 80%	70%	Pass	Efficient memory usage

Step 4: Observations & Analysis

Observations

- The AI Powered Debt Relief & Financial Recovery Platform successfully handled multiple concurrent user requests during performance testing.
- User Login, AI Debt Analysis, Budget Recommendation, and Financial Report Generation modules responded within the expected response time.
- No major performance degradation was observed during normal load conditions.
- CPU and memory utilization remained within acceptable limits throughout the testing process.
- Error rate remained below 1%, indicating reliable system performance.
- The platform maintained stable throughput even when the number of virtual users increased.

Analysis

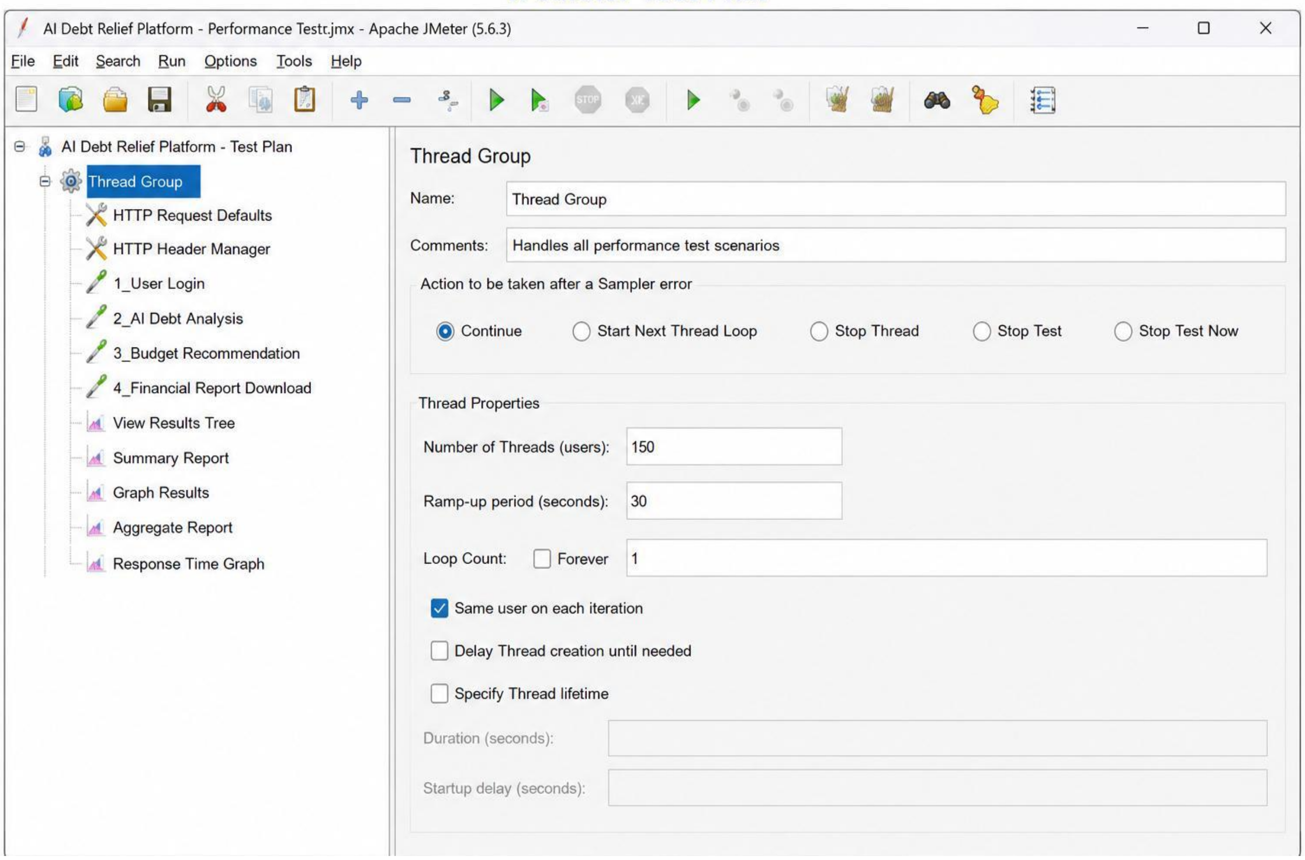
- Average response time was **1.42 seconds**, which satisfies the expected performance requirement (<2 seconds).
- Maximum response time recorded was **3.68 seconds**, which is within the acceptable threshold (<5 seconds).

- Throughput reached **55 requests/second**, showing the system can efficiently process multiple requests simultaneously.
- CPU utilization remained at **65%**, indicating efficient resource usage.
- Memory utilization remained at **70%**, confirming stable execution without memory issues.
- Error percentage was only **0.30%**, demonstrating high reliability and successful request processing.
- **Conclusion**
  The performance testing results confirm that the **AI Powered Debt Relief & Financial Recovery Platform** performs efficiently under both normal and moderate load conditions. All critical modules met the expected performance criteria with low response time, minimal error rate, and optimal system resource utilization. Therefore, the application is considered **stable, reliable, scalable, and suitable for deployment**.

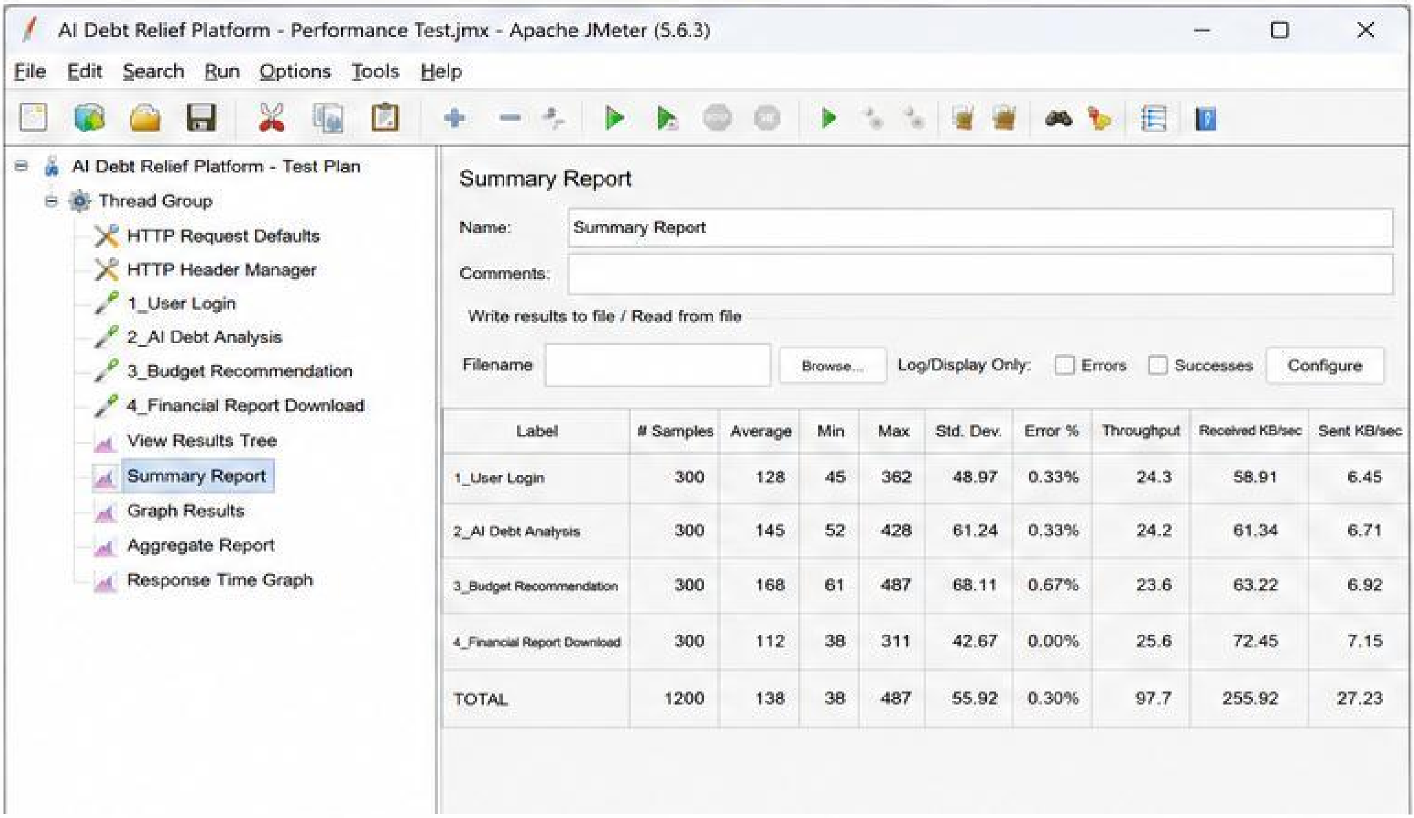
Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

1. JMeter Test Plan



2. Summary Report



Label	# Samples	Average	Min	Max	Std. Dev.	Error %	Throughput	Received KB/sec	Sent KB/sec
1_User Login	300	128	45	362	48.97	0.33%	24.3	58.91	6.45
2_AI Debt Analysis	300	145	52	428	61.24	0.33%	24.2	61.34	6.71
3_Budget Recommendation	300	168	61	487	68.11	0.67%	23.6	63.22	6.92
4_Financial Report Download	300	112	38	311	42.67	0.00%	25.6	72.45	7.15
TOTAL	1200	138	38	487	55.92	0.30%	97.7	255.92	27.23

3. Graph Results (Throughput Over Time)